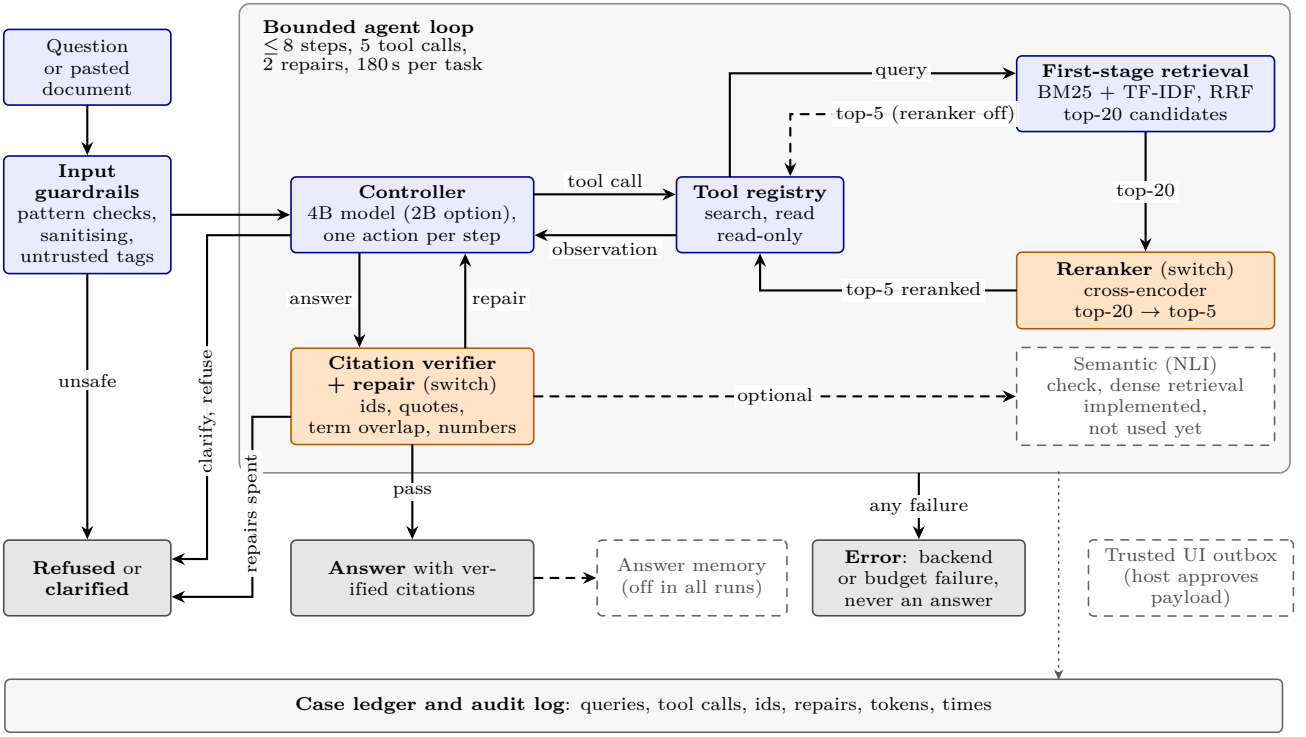

Bounded agent loop
≤ 8 steps, 5 tool calls,
2 repairs, 180 s per task
Question or pasted document
Input guardrails
pattern checks, sanitising, untrusted tags
unsafe
clarify, refuse
Refused or clarified
Controller
4B model (2B option), one action per step
tool call
observation
answer
repair
repairs spent
Tool registry
search, read
read-only
query
top-5 (reranker off)
First-stage retrieval
BM25 + TF-IDF, RRF
top-20 candidates
top-20
Reranker (switch)
cross-encoder
top-20 → top-5
top-5 reranked
Citation verifier + repair (switch)
ids, quotes,
term overlap, numbers
optional
Semantic (NLI) check, dense retrieval implemented, not used yet
pass
Answer with verified citations
Answer memory (off in all runs)
any failure
Error: backend or budget failure, never an answer
Trusted UI outbox (host approves payload)
Case ledger and audit log: queries, tool calls, ids, repairs, tokens, times